Exam Problem 4 • Industrial Manufacturing

CEMED – Industrial Manufacturing Accounting Cycle & Trade Discount Analysis

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CERÁMICAS DEL MEDITERRÁNEO (CEMED) was created 1/1/2016 by 3 partners contributing 7 mu each in cash (total 21 mu).

• 1/2/2016: bought land for 10 mu (50% cash, 50% payable 1 year later); factory building for 6 mu in cash financed by a 10-year mortgage at 5% (20-year linear life, returns 10% principal each 31 Dec + interest).

• 1/4/2016: borrowed 5 mu at 6% from bank, returning 20% each semester for 2.5 years (1st repayment 1/10/2016, 2nd 1/4/2017). Purchased machinery for 3 mu (60-day payment delay; 10-year linear life).

• 1/7/2016: bought 150 tons raw material at 0.03 mu/ton (half used in sales; 60% paid at 90 days, 40% pending).

• Sales: 50 mu (70% collected). Personnel & SS: 0.8 mu paid in full. Tax: 30% paid on 15/01/2017. Reserves: 40% of net profits, 60% dividends payable on 31/12/2016.

1. Prepare 2016 Income Statement and Balance Sheet at 24:00 on 31/12/2016.

2. Analyze impact of 5% raw material purchase discount in exchange for paying the 60% in cash.

EXAM DAY STRATEGY: SEMIANNUAL VS ANNUAL LOAN CALCULATIONS

- **Factory Building (11 months):** $D = \frac{6.00}{20} \times \frac{11}{12} = 0.275$ mu. Mortgage interest: $6.00 \times 5\% \times \frac{11}{12} = 0.275$ mu. Principal returned on 31 Dec $= 0.60$ mu.
- **Bank Loan (Repayments & Accruals):** On 1 Oct returned 1.00 mu principal + paid 0.15 mu interest ($5 \times 6\% \times 0.5$). For Q4 (1 Oct to 31 Dec), remaining principal is 4.00 mu, accruing $4 \times 6\% \times \frac{3}{12} = \mathbf{0.06}$ mu unpaid interest payable in April 2017!
- **Machinery (9 months):** $D = \frac{3.00}{10} \times \frac{9}{12} = 0.225$ mu.

Statements & Balanced Verification

COGS $= 2.25$ mu. Operating Expenses $= 2.25 + 0.80 + 0.50$ (dep) $= 3.55$ mu. EBIT $= 50.00 - 3.55 =$ **46.45** mu. Financial Interest $= 0.275 + 0.210 =$ **0.485** mu. EBT $= 45.965$ mu. Tax (30%) $= 13.7895$ mu. Net Income $=$ **32.1755** mu (Reserves: 12.8702 mu, Dividends: 19.3053 mu).

ASSETS at 31/12/2016	Amount (mu)	LIABILITIES & EQUITY at 31/12/2016	Amount (mu)
• Land (Non-depreciable)	10.0000	• Subscribed Share Capital	21.0000
• Factory Building (Net: 6.00 - 0.275)	5.7250	• Retained Reserves (40% of Net Income)	12.8702
• Machinery (Net: 3.00 - 0.225)	2.7750	**TOTAL EQUITY**	**33.8702**
TOTAL NON-CURRENT ASSETS	**18.5000**	**• Long-term Mortgage (4.80) & Loan (2.00)**	**6.8000**
• Inventories (Raw Materials)	2.2500	• Land debt due 1/2/2017	5.0000
• Accounts Receivable (30% of sales)	15.0000	• Short-term Mortgage (0.60) & Loan (2.00)	2.6000
• Cash & Liquid Treasury	28.1697	• Accrued bank loan interest	0.0600
		• Accounts Payable (Raw materials 40%)	1.8000
		• Taxes Payable (due 15/01/2017)	13.7895
TOTAL ASSETS	**63.9197**	**TOTAL LIABILITIES & EQUITY**	**63.9197**

Cash derivation (31/12/2016): inflows $21 + 6 + 5 + 35 = 67.0000$ mu; outflows $5 + 6 + 3 + 2.7 + 0.8 + 0.6 + 0.275 + 1.0 + 0.15 + 19.3053 = 38.8303$ mu (land down payment, building, machinery, 60% of raw materials, personnel, mortgage principal and interest, bank loan principal and first-semester interest, dividends); cash $= 67.0000 - 38.8303 =$ **28.1697** mu.

TRADE DISCOUNT NEGOTIATION VERDICT (QUESTION 2)

A 5% discount on purchases reduces material cost by 0.225 mu. Because half was consumed, **COGS decreases by** 0.1125 mu, expanding EBIT by $+0.1125$ mu and Net Profit by $+0.1125 \times 0.70 =$ $+$**0.07875** mu. Profit is strictly higher: the income statement does not record the time value of paying the 60% earlier, and both payment dates fall within 2016. The balance sheet also changes: inventories -0.1125 mu, accounts payable -0.09 mu, taxes payable $+0.03375$ mu, reserves $+0.0315$ mu, and year-end cash $+0.08775$ mu ($= 0.135$ saved on the supplier payment -0.04725 extra dividend), so that $\Delta\text{Assets} = \Delta(\text{Liabilities} + \text{Equity}) = -0.02475$ mu.

Four-Phase Audit Trail (journal entries and ledger)

Phase 1: Hypotheses. Calendar year 2016; partners' 21 paid in cash. The mortgage (6 at 5%) funds the building and its 10% principal plus annual interest are paid each 31 Dec. The bank loan (5 at 6%) is repaid 20% each semester from 1 Oct 2016, with interest on the amount actually due; interest for 1 Oct to 31 Dec ($4 \times 6\% \times 3/12 = 0.06$) is accrued. Machinery (3) is paid on 31 May. Raw materials (4.5) are bought on 1 Jul, half consumed; 60% paid on 1 Oct, 40% open at year end. Tax (30%) is payable 15 Jan 2017; dividends (60% of net income) are paid on 31 Dec and 40% is kept as reserves.

Phase 2: Definitions. $A = L + E$ with $E =$ Share capital + Reserves and Reserves $= 0.40 \times$ Net income. Land is not depreciated; building $6/20 \times 11/12$; machinery $3/10 \times 9/12$. Debt due within 12 months is current (mortgage 0.6; bank loan 2.0).

Phase 3: Journal entries, Part 1 (m.u.). Each entry was balanced in Python with exact fractions (debits = credits).

#	Event	Debit	Credit
1	1 Jan: partners' capital 3 x 7	Cash 21.0000	Share capital 21.0000
2	1 Feb: land 10 (50% cash, 50% due 1/2/2017)	Land 10.0000	Cash 5.0000 Land payable 5.0000
3	1 Feb: factory building 6 financed by mortgage at 5%	Building 6.0000	Mortgage 6.0000
4	1 Apr: bank loan 5 at 6%	Cash 5.0000	Bank loan 5.0000
5	1 Apr: machinery 3, paid after 60 days (31 May)	Machinery 3.0000	Cash 3.0000
6	1 Jul: raw materials 150 t x 0.03 = 4.5	Inventory 4.5000	Suppliers payable 4.5000
7	Payment of 60% of the raw materials at 90 days (1 Oct)	Suppliers payable 2.7000	Cash 2.7000
8	Raw materials consumed: half	COGS 2.2500	Inventory 2.2500
9	Sales 50: 70% collected, 30% open	Cash 35.0000 Accounts receivable 15.0000	Sales 50.0000
10	Personnel and Social Security paid	Personnel 0.8000	Cash 0.8000
11	Depreciation building 6/20 x 11/12	Depreciation 0.2750	Accum. dep. building 0.2750
12	Depreciation machinery 3/10 x 9/12	Depreciation 0.2250	Accum. dep. machinery 0.2250
13	31 Dec: mortgage interest 6 x 5% x 11/12 and principal 10% x 6 paid	Interest 0.2750 Mortgage 0.6000	Cash 0.8750
14	1 Oct: loan principal 20% x 5 and interest 5 x 6% x 6/12 paid	Bank loan 1.0000 Interest 0.1500	Cash 1.1500
15	31 Dec: accrued interest 4 x 6% x 3/12 (paid 2/4/2017)	Interest 0.0600	Accrued interest 0.0600
16	31 Dec: corporate tax 30% (paid 15/1/2017)	Tax expense 13.7895	Taxes payable 13.7895
17	31 Dec: dividends 60% of net income paid	Dividends 19.3053	Cash 19.3053

Closing ledger account (31/12/2016)	Balance (m.u.)
Cash	28.1697 Dr
Accounts receivable	15.0000 Dr
Inventory	2.2500 Dr
Land	10.0000 Dr
Building	6.0000 Dr
Accum. dep. building	0.2750 Cr
Machinery	3.0000 Dr
Accum. dep. machinery	0.2250 Cr
Land payable	5.0000 Cr
Mortgage	5.4000 Cr
Bank loan	4.0000 Cr
Accrued interest	0.0600 Cr
Suppliers payable	1.8000 Cr
Taxes payable	13.7895 Cr
Share capital	21.0000 Cr

Statement lines: $\text{EBIT} = 50 - 2.25 - 0.8 - 0.5 = 46.45$; interest $= 0.275 + 0.15 + 0.06 = 0.485$; $\text{EBT} = 45.965$; tax $= 0.30 \times 45.965 = 13.7895$; net income $= 32.1755$; dividends $= 0.60 \times 32.1755 = 19.3053$; reserves $= 12.8702$ (closed from income and dividends, so not a ledger line above). Assets $18.5 + 45.4197 = 63.9197$; liabilities $5 + 5.4 + 4.0 + 0.06 + 1.8 + 13.7895 = 30.0495$ plus equity $21 + 12.8702 = 33.8702$.

Part 2 (5% discount, 60% paid in cash on 1 Jul): purchase price $4.5 \times 0.95 = 4.275$; payment 2.565. Changes with respect to Part 1:

Item	Change vs Part 1 (m.u.)
Net income	+0.07875
Cost of goods sold	-0.1125
Inventory	-0.1125
Cash	+0.08775
Suppliers payable	-0.09
Taxes payable	+0.03375
Total assets	-0.02475
Total liabilities + equity	-0.02475

Cash: smaller supplier payment $2.70 - 2.565 = +0.135$, minus the extra dividend 0.04725 (60% of 0.07875) gives $+0.08775$. Balance check: $\Delta A = -0.1125 + 0.08775 = -0.02475 = -0.09 + 0.03375 + 0.0315 = \Delta(L + E)$.

Phase 4: Units and sanity checks. Currency: m.u.; $A = L + E$: $63.9197 = 30.0495 + 33.8702$ (Part 1) and $63.89495 = 63.89495$ (Part 2). Cash identity Part 1: $67.0000 - 38.8303 = 28.1697$. Limit check: with a 0% discount every change vanishes; net income rises by the after-tax saving $0.1125 \times (1 - 0.30) = 0.07875$, so profit is higher with the discount. No official numerical key is available in `sources/` (only the statement PDF).

Study notes written from the official course material (T3_Additional_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-extra-exam-problems.html